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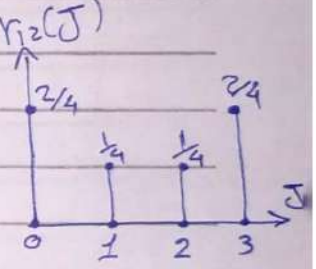
Sheet 3

#3.1 Convolution → How To input interacts with system To Produce o/p.
Correlation → measured degree of Similarity Between 2 Signals.

#3.2 Cross-Correlation → measured for Between 2 different Signals.
Correlation Theorem → measured degree of Similarity Between 2 Signals.
Correlation Coefficient → $\rho_{12} = \frac{r_{12}}{\frac{1}{N_{new}}\sqrt{\sum x_1^2(n) \cdot \sum x_2^2(n)}}$

#3.3 Circular Correlation
↳ {1,1,0,1} {1,0,0,1}
$N_{new} = MAX(N_1, N_2) = 4$

X(n)	1	1	0	1	J	$r_{12}(J)$	$\rho_{12}(J)$
	1	0	0	1	0	2/4	0.816
X(n+j)	0	0	1	1	1	1/4	0.408
	0	1	1	0	2	1/4	0.408
	1	1	0	0	3	2/4	0.816



$r_{12}(J) = \frac{1}{N_{new}} \sum_{n=0}^{N-1} x(n)\,X(n+j)$ $\rho_{12}(J) = \frac{r_{12}(J)}{\frac{1}{N_{new}}\sqrt{\sum_{n=0}^{N} x_1^2(n) \cdot \sum_{n=0}^{N} x_2^2(n)}} = \frac{r_{12}(J)}{0.612}$

#3.4 Linear Correlation
↳ $N_{new} = N_1 + N_2 - 1 = 4+4-1 = 7$ $\rho_{12}(J) = \frac{r_{12}(J)}{0.408}$

X(n)	1	1	0	1	0	0	0	J	$r_{12}(J)$	$\rho_{12}(J)$
	1	0	0	1	0	0	0	0	2/6	0.816
	0	0	1	0	0	0	1	1	Zero	Zero
X(n+j)	0	1	0	0	0	1	0	2	1/6	0.408
	1	0	0	0	1	0	0	3	1/6	0.408
	0	0	0	1	0	0	1	4	1/6	0.408
	0	0	1	0	0	1	0	5	Zero	Zero
	0	1	0	0	1	0	0	6	1/6	0.408

CamScanner

#3.5 {1.5, 2, 1.5, 2.25} {0, 0.33, 0.67, 1}

ⓐ Auto Correlation

Nnew = 5 Nnew = 4

X(n)	1.5	2	1.5	2	2.25	J	$r_{11}(J)$	$\rho_{11}(J)$
X(n+J)	1.5	2	1.5	2	2.25	0	3.5125	1
	2	1.5	2	2.25	1.5	1	3.375	0.960
	1.5	2	2.25	1.5	2	2	3.425	0.975
	2	2.25	1.5	2	1.5	3	3.425	0.975
	2.25	1.5	2	1.5	2	4	3.375	0.960

وتستفاد التانية نفس الحل

$$r_{11}(J) = \frac{1}{N_{new}} \sum_{n=0}^{N-1} X(n)\,X(n+J) \qquad \rho_{11}(J) = \frac{r_{11}(J)}{\frac{1}{N_{new}} \sum X_1^2(n)} = \frac{r_{11}(J)}{3.5125}$$

ⓑ Linear Nnew = 5+4-1 = 8

X(n)	1.5	2	1.5	2	2.5	0	0	0	J	$r_{12}(J)$	$\rho_{12}(J)$
X(n+j)	0	0.33	0.67	1	0	0	0	0	0	0.458	0.678
	0.33	0.67	1	0	0	0	0	0	1	0.416	0.617
	0.67	1	0	0	0	0	0	0.33	2	0.375	0.556
	1	0	0	0	0	0	0.33	0.67	3	0.187	0.277
	0	0	0	0	0	0.33	0.67	1	4	Zero	Zero
	0	0	0	0	0.33	0.67	1	0	5	0.103	0.152
	0	0	0	0.33	0.67	1	0	0	6	0.291	0.432
	0	0	0.33	0.67	1	0	0	0	7	0.541	0.802

$$r_{12}(J) = \frac{1}{N_{new}} \sum_{n=0}^{N-1} X(n)\,X(n+j)$$

$$\rho_{12}(J) = \frac{r_{12}(J)}{\frac{1}{N_{new}} \sqrt{\sum_{n=0}^{N} X_1^2(n) \cdot X_2^2(n)}} = \frac{r_{12}(J)}{0.6755}$$

نفس الحل في نقطة ⓒ

#3.6 input {0, 2.5, 5, 0}
impulse Response {0, 0.899, 0.990, 0.991}

(a) Linear Convolution

$N_{new} = N_1 + N_2 - 1 = 4 + 4 - 1 = 7$

x(m)	0	0	0	0	2.5	5	0	n	y(n)
	0.991	0.990	0.899	0	0	0	0	0	Zero
	0	0.991	0.990	0.899	0	0	0	1	Zero
h(n-m)	0	0	0.991	0.990	0.899	0	0	2	2.247
	0	0	0	0.991	0.990	0.899	0	3	6.97
	0	0	0	0	0.991	0.990	0.899	4	7.427
	0	0	0	0	0	0.991	0.990	5	4.955
	0	0	0	0	0	0	0.991	6	Zero

(b) Circle Convolution

$x(m) = \{0, 2.5, 5, 0\}$

$h(-m) = \{0, 0.991, 0.990, 0.899\}$

x(m)	0	2.5	5	0	n	y(n)
	0	0.991	0.990	0.899	0	7.427
h(n-m)	0.899	0	0.991	0.990	1	4.955
	0.990	0.899	0	0.991	2	2.247
	0.991	0.990	0.899	0	3	6.97